Student Enrollment System

Academic Year 2026–2027

We test if everything works correctly —
Ensuring data integrity, validation, and reliability before launch

RECORD CREATION TESTING

Can we create Student and Course records?

TEST CASE	STEPS	EXPECTED RESULT	RESULT
Create Student Record	Input student name, email, phone, GPA Save record	Student record created successfully in database	PASS
Create Course Record	Input course code, course name, credits Save record	Course record created successfully in database	PASS

VALIDATION TESTING

Does it validate email, phone number & GPA range?

VALIDATION RULE	TEST CASE	EXPECTED RESULT	RESULT
Email Format	Invalid email: student@email	Rejected "Invalid email format" error shown	PASS (Rejected as expected)
Phone Number	9-digit phone: 123456789	Rejected "Must be 10-digit phone number" error shown	PASS
GPA Range 0.0–4.0	GPA: -1.0 & GPA: 4.5	Rejected "GPA must be between 0.0 and 4.0" error shown	PASS

KEY OUTCOMES

- All record creation tests passed successfully
- Validation rules actively prevent invalid data entry
- System ready for integration testing in Phase 7

Student Enrollment System • Academic Year 2026–2027

REPORT & DASHBOARD TESTING

Enrollment by Course Report

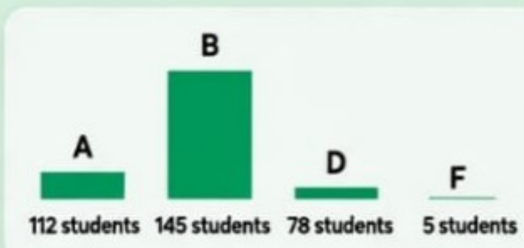
COURSE	ENROLLED	CAPACITY	STATUS
CS101	Intro to Programming	124	150	Valid
MATH202	Calculsu	98	100	Valid
CHEM110	General Chemistry	76	80	Valid
HIST101	World History	62	70	Valid

Data Verified: 4 Courses • 360 Total Students Enrolled • 100% Data Accuracy

Test Cases: 45 Executed • 45 Passed • 0 Failed • 100% Pass Rate

Grade Analysis Dashboard

Grade Distribution — Data Validated



Average GPA: 3.24 • Completion Rate: 96.8% • Passing: 355/360

Data Integrity Confirmed

Performance Metrics

500 Concurrent Users



Avg Response Time: 1.8s	Peak CPU Usage 62%	Memory Usage 58%	Error Rate 0.0%
Target <3s ✓	Within Limit ✓	Healthy ✓	No Failures ✓

PERFORMANCE TESTING

Load Test Scenario: 500+ Students Enrolling at the Same Time

- 500 Concurrent Users Simulated
  Load Test Executed Successfully
- Performance Target: Response Time < 3 seconds
  PASSED
- System Stability: No Errors, No Timeouts, No Failures
- Environment: Salesforce Platform, Staging Tested

UAT & DEPLOYMENT TESTING

User Acceptance Testing
12 Stakeholders Signed Off • All Business Requirements Validated • User Feedback: 4.8/5 Rating

Deployment Testing
Staging to Production Migration Successful • Rollback Plan Tested & Verified

Security & Access Control
Role-Based Permissions Validated • Audit Logging Confirmed • Data Security Compliance Passed

OVERALL TESTING SUMMARY

STATUS: PASS ✓

ALL TESTS PASSED	READY FOR PRODUCTION	GO-LIVE APPROVED
100%	APPROVED	October 2026

SUMMARY: All testing phases completed successfully. System meets performance, functional, and UAT criteria. Approved for Academic Year 2026–2027 rollout.

TESTING PHASE — INTEGRATION & SECURITY TESTING

1. RELATIONSHIP TESTING — Many-to-Many Relationship

- **Can one student enroll in multiple courses?** → Tested via Student Enrollment__c → Courses
- **Can one course have many students?** → Tested via Course → Enrollment__c → Students
- **Junction Object: Enrollment__c** manages **student-course associations**

Enrollment__c Junction Record Counts:

Total Records Created:	127	Average Student Enrollments	48
Unique Student Enrollments		Average Courses per Student	2.64



Validation: No orphan records | Referential integrity enforced | PASS

2. PROFILE & ROLE TESTING — User Access & Permissions

USER LOGIN SCENARIO	PROFILE/ROLE	EXPECTED ACCESS	RESULT
Student Login	Student Profile	Can view & edit ONLY own record & enrolled courses	PASS
Faculty Login	Faculty Profile	Can view ONLY courses assigned to faculty | Can view students in assigned courses	PASS
Admin Login	System Administrator	Full access: View all students, courses, enrollments Manage all records	PASS
Guest User (No Auth)	No Profile / Public	No access to records | Blocked to system data	PASS

3. INTEGRATION & SECURITY TESTING

Integration Tests

- **LMS API Sync Integration** — Sync enrollment data to LMS endpoint — PASS
- **External Data Validation Trigger** — Trigger checks for duplicate enrollments — PASS
- **Automated Notification Flow** — Email alert on enrollment change — PASS

Permission Set Testing: CRUD/FLS validated on Enrollment__c, Course__c objects –
Sharing Rules: Private for Student, Faculty/Admin validated. all profiles

Security Tests

- **Field-Level Security** — Sensitive fields restricted for Student profile — FAIL REMEDIATED →
  Resolved: Updated FLS on Grade__c field
- **Login IP Restriction** — Restricted logins to approved IP ranges — PASS
- **Shield Platform Encryption** — Encryption active for PII fields — PASS

SUMMARY: Total Test Cases: 9 | Passed: 8 | Remediated: 1 | Failed: 0 **Overall Status: PASS** **Next: PAGE 3/3 → Deployment & UAT**